# Computational Logic

## Introduction

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# Introduction

# Who am I?

# Entry poll

- Have you written a proof by induction? (yes / seen it / no)
- Do you know what a truth table is?
- Can you read $\forall$ and $\exists$ comfortably?
- Is this course something you chose, or something you have to pass?

# Logical problems and what the class is about

# Ball & Bat

## Shane Frederick, 2005

- A ball and a bat together cost $1.10.
- The bat costs $1 more than the ball.
- How much does the ball cost?



# Take home message

- Everyone who says 10¢ is certain while saying it
- Being wrong felt exactly like being right
- Intuition doesn't report its own failures
- So we need arguments others can check → proof, from week 2

# Wason’s selection taks

## Check the following rule:

You are shown four cards placed on a table, each with a number on one side and a color on the other. The visible faces of the cards are 3, 8, blue, and red.

Which card(s) must you turn over to test that if a card shows an even number on one face, its opposite face is blue?



# Take home message

- Answer: 8 and Red — about 1 in 10 get it
- We over-read "if P then Q" as "P if and only if Q"
- We hunt for confirmations; only falsifiers count
- One connective is doing all the damage: →
- → truth table for →, week 2

# Wason’s deontic version

- Beer, cola, 22, 16 → turn beer and 16
- About 3 in 4 get it. Same logical form
- Only the story changed
- Intuition tracks content, not structure
- → logical form vs. what people (and machines) compute, week 8
- → modal logic, week9

# An Example from Hector Levesque

- Jack is looking at Anne, but Anne is looking at George. Jack is married, but George is not. Is a married person looking at an unmarried person?

- (a) Yes
  (b) No
  (c) Not enough information to determine

# Take home message

- Answer: yes
- Anne is married, or she isn't — both cases give yes
- Missing information ≠ missing answer
- Reasoning by cases → $\vee$-elimination, week 2

# A syllogistic reasoning

Some zookeepers are pacifists.
No pacifists are troglodytes.

Does it follow that some zookeepers are not troglodytes?

# Take home message

- Valid — but why?
- Swap in primes and even numbers: just as good
- Validity is a property of form, not subject matter
- The form contains some and no — quantifiers
- → first-order logic, weeks 5–8

# Another syllogism

Some zookeepers are pacifists.

Some pacifists are troglodytes.

Does it follow that some zookeepers are troglodytes?

# Take home message

- Some pacifists are troglodytes — one word changed
- Reads the same. Feels the same. Invalid
- Refuted by building a situation, not by arguing
- Premises true + conclusion false = counter-model
- → proof or counter-model — know which you need (weeks 3, 5)

# Prisoners' hat riddle



# Take home message

- Ten prisoners, one line, each sees only forward
- Nine must be right — guessing gives you 1 in 1024
- The one at the back sends one bit: parity
- "Black" if he sees an odd number of black hats
- Everyone else counts, subtracts, and knows
- → XOR, week 2 · knowledge and announcement, weeks 9–11

# Do we have two volunteers to enact a puzzle?

# Muddy Children Puzzle

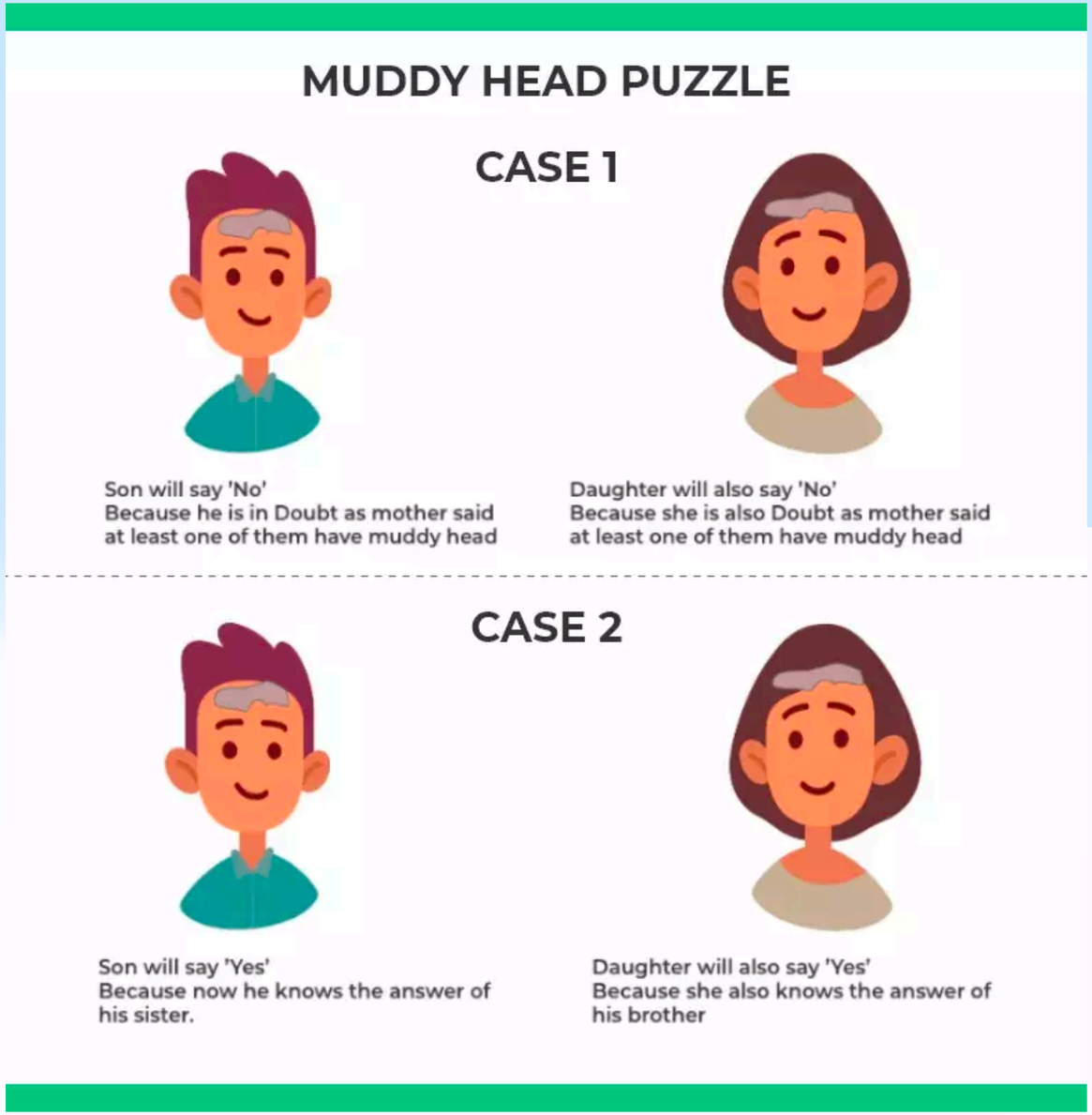
A mother tells her two children, boys and girls, to play without getting dirty.

However, while playing, both children get mud on their foreheads.

The mother says, “At least one of you has a muddy forehead.” She then asks the children to answer “Yes” or “No” to the question: “Do you know whether you have a muddy forehead?”

The mother asks this question twice.

What will the children answer each time this question is asked, assuming that a child can see whether their sibling has a muddy forehead but cannot see their forehead? Assume that both children are honest and that the children answer each question simultaneously.



# Take home message

- Round 1: "no". Round 2: "yes". What changed?
- Nobody looked in a mirror; no new fact about mud
- Each child learned from the other's silence
- She doesn't know → she must see mud on me
- Knowledge about knowledge — unsayable so far
- → modal logic, weeks 9–11

# Where these go

- Cards → the conditional → weeks 2–4
- Zookeepers, Jack & Anne → quantifiers, cases → weeks 5–8
- Hats, muddy children → knowledge about knowledge → weeks 9–11
- Bat & ball → all of it: answers you can check
- You won't be cleverer in December — you'll be less dependent on being clever

# Excercise

Go to your favorite LLM and try to break it on a logical problem. Share your results with us all.

# Take home message

- How do you know it’s right ot wrong?
- A system that is usually right is not the same as a system you can check.

# So, what is logic, after all?



# Four questions

- Form — when does a conclusion follow from premises, whatever the words mean?
- Truth — what makes a statement true in a situation?
- Proof — what is an argument you can check without trusting anyone?
- Computation — which of these questions can a machine answer, and how fast?
- Logic is the study of these four and, more interestingly, of how they fit together.

# Form — why a machine can check an argument

- The zookeepers argument was valid without using anything about zookeepers
- Validity depends on shape, not on subject matter
- So it can be checked by something that understands nothing
- That is the whole reason reasoning can be mechanized
- Your type checker, your linter, your compiler's optimizer: all of them check form
- → week 2: we build a formal language whose shapes are explicit

# Truth — a model is a data structure

- A model is a situation: some objects, some relations that hold between them
- A formula is true in a model — truth is relative to a situation, and precisely so
- A database is a model. A query is a formula. Evaluating a query is checking truth in a model
- Codd's relational algebra and first-order logic have the same expressive power
- Same story for hardware: a circuit state is an assignment; a spec is a formula
- → weeks 5–8: models, satisfaction, and what first-order logic cannot say

# Proof — an argument that survives distrust

- A proof is a finite, syntactic object — checkable step by step, no appeal to meaning
- Checking a proof is easy. Finding one is hard
- That gap has a name you already know: P vs NP
- Proofs are programs: checking a proof ≈ checking a type
- seL4: an operating-system kernel with a machine-checked correctness proof
- → weeks 2–3: natural deduction, soundness, completeness

# Computation — the boundary, and the tools

- Propositional validity: decidable — but coNP-complete, so no known fast algorithm in general
- First-order validity: undecidable — Church and Turing, 1936. No algorithm. Ever
- And yet: SAT solvers handle industrial problems with millions of clauses
- The gap between worst case and practice is where the engineering lives
- → week 4 lab, §1.6: you run a SAT solver yourself. Bring a laptop

# You are already using this

- Circuits are formulas. Gates, Boolean algebra, minimization — that's normal forms and functional completeness
- Equivalence checking. "Does my optimized netlist compute the same function as the RTL?" = Is this a tautology? = SAT
- Chip verification. Assertion languages are logics; model checking won the 2007 Turing Award
- Package managers. Dependency resolution is a SAT problem, and your distro solves it with a SAT solver
- Cloud policy. Access-control rules are formulas; "can anyone outside reach this bucket?" is a validity question
- Compilers, drivers, flight control. Verified toolchains and static analyzers, all built on this
- Intel, 1994: a floating-point division bug, a ~$475M charge — and formal verification became standard practice

# So what this course actually gives you

- Not "how to think" — you already think
- A language precise enough that a machine can check what you meant
- The ability to tell a proof from a plausible story
- A sense of where the walls are: what's decidable, what's tractable, what's neither
- Four names to recognise later: Boolean algebra · relational queries · type systems · SAT
- All four have the same ancestor. You're about to meet it

# Class organization

# How this class runs

- One 4-hour session a week, lecture and exercises interleaved
- Before class: study the sections listed for that week
- In class: I develop the material; you solve problems
- Ask in class. If it's unclear to you, it's unclear to others

# What is obligatory? What is graded?

- **Obligatory, not graded**
  - Three homework assignments — must be submitted by the deadline to sit the exam
  - Model answers discussed in class
  - Trial intermediate exam, week 7
  - Trial final exam, week 12
- **Graded**
  - One written exam, 2 hours, closed-book
  - Nothing else. No midterm, no graded coursework
  - Marks out of 30

# The exam

- 2 hours, closed-book, marked out of 30
- Three kinds of exercise:
- Formalize — informal statement → formula
- Semantics — satisfiable? valid? a consequence? Show a model or a counter-model
- Prove — build a derivation, or a short metatheoretical argument
- Plus precise statement of definitions and results
- A right answer with no valid argument gets partial credit

# The books

- Main: Huth & Ryan, Logic in Computer Science, 2nd ed. — section numbers in the schedule refer to this
- Free alternative: Zach et al., Sets, Logic, Computation (Open Logic Project) — open access, if you'd rather not buy a book
- Optional: van Benthem, Modal Logic for Open Minds — further reading for weeks 9–11
- Optional: Gonczarowski & Nisan, Mathematical Logic through Python — the material by implementing it
- Optional: Logicomix — read it after today. Not examined

# The plan

- Weeks 2–4 · Propositional logic — "how to win a million dollars with logic", ending in a SAT-solver lab
- Weeks 5–8 · First-order logic — quantifiers, non-computability, expressiveness and definability
- Weeks 9–11 · Modal logic — knowledge, time, agency
- Week 12 · Q&A and trial final exam
- Weeks 13–15 · in reserve — used only if a session has to be canceled
- Full week-by-week schedule with readings: on Moodle

- Week 1 (14.09): Intro (try to read Logicomix after)
- Week 2 (21.09): How to Win a Million Dollars with Logic? Part 1:The language and what it means. (Study Sections 1.1, 1.3, 1.4.1-1.4.2 of the book)
- Week 3: (28.09) How to win a million dollars with logic? Part 2: Proof - and why it matches meaning. (Study 1.2, 1.4.3-1.4.4)
- Week 4: (05.10) How to win a million dollars with logic? Part 3: Normal forms and solvers. Lab, BY laptops (Study 1.5, 1.6)
- Homework 1.
- Week 5: (12.10) FO and non-computability (Study 2.1-2.2)
- Week 6: (19.10) FO and non-computability (Study 2.3-2.4)
- Week 7: (26.10) FO and non-computability (Study 2.5) + Trial intermediate exam.
- Homework 2.
- Week 8: (02.11) Expressiveness, Definability, and Generalized Quantifier Theory (Study 2.6-27, and the book chapter on Moodle)
- Week 9: (09.11) Modal Logic and Agency 1 (Study 5.2-5.3)
- Week 10: (16.11) Modal  Logic and Agency 2 (Study 5.4-5.5)
- Week 11: (23.11) Modal  Logic and Agency 3.
- Homework 3.
- Week 12: (30.11) Q&A + Trial final exam.

# Practicalities

- 6 CFU · 48 contact hours · ~102 hours of your own work
- Taught in English
- Moodle: didatticaonline.unitn.it/dol/course/view.php?id=44770
- Office hours by appointment — write to me
- Use the forum: fastest answer, and it tells me what to re-explain
- Accommodations (disability / DSA): contact the inclusion services and me early
- AI assistants: fine for studying — but the exam is handwritten and unassisted

# Questions?